```
!pip install -q ultralytics opencv-python matplotlib pandas
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from ultralytics import YOLO

print("Libraries imported successfully!")
```

```
Creating new Ultralytics Settings v0.0.8 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For
Libraries imported successfully!
```

```
# Load YOLOv8 pre-trained model
model = YOLO("yolov8n.pt")

print("YOLOv8 model loaded successfully!")
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt to 'yolo
YOLOv8 model loaded successfully!
```

```
from google.colab import files

uploaded = files.upload()

image_path = list(uploaded.keys())[0]

print("Uploaded file:", image_path)
```

Choose Files bmw m4.jpg

bmw m4.jpg(image/jpeg) - 8001429 bytes, last modified: 9/24/2026 - 100% done

```
Saving bmw m4.jpg to bmw m4.jpg
Uploaded file: bmw m4.jpg
```

```
image = cv2.imread(image_path)

# Convert BGR to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(10, 7))
plt.imshow(image_rgb)
plt.axis("off")
plt.title("Original Image")
plt.show()
```




```
results = model(image_path)

print("Object detection completed!")
```

```
image 1/1 /content/bmw m4.jpg: 448x640 1 car, 298.7ms
Speed: 26.3ms preprocess, 298.7ms inference, 27.8ms postprocess per image at shape (1, 3, 448,
Object detection completed!
```

```
result = results[0]

detected_image = result.plot()

detected_image_rgb = cv2.cvtColor(
    detected_image,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(12, 8))
plt.imshow(detected_image_rgb)
plt.axis("off")
plt.title("YOLOv8 Object Detection")
plt.show()
```




```
boxes = result.boxes

print("Detected Objects:\n")

for box in boxes:
    class_id = int(box.cls[0])
    confidence = float(box.conf[0])

    class_name = model.names[class_id]

    print(
        f"Object: {class_name} | "
        f"Confidence: {confidence:.2f}"
    )
```

```
Detected Objects:

Object: car | Confidence: 0.79
```

```
print("Bounding Box Information:\n")

for box in boxes:

    class_id = int(box.cls[0])
    confidence = float(box.conf[0])

    x1, y1, x2, y2 = box.xyxy[0].tolist()

    class_name = model.names[class_id]
```

```
    print(f"Object: {class_name}")
    print(f"Confidence: {confidence:.2f}")
    print(
        f"Bounding Box: "
        f"({x1:.0f}, {y1:.0f}, {x2:.0f}, {y2:.0f})"
    )
    print("-" * 40)
```

```
Bounding Box Information:

Object: car
Confidence: 0.79
Bounding Box: (1786, 1990, 6286, 4251)
----------------------------------------
```

```
object_counts = {}

for box in boxes:

    class_id = int(box.cls[0])
    class_name = model.names[class_id]

    if class_name not in object_counts:
        object_counts[class_name] = 0

    object_counts[class_name] += 1

print("Object Counts:")

for obj, count in object_counts.items():
    print(f"{obj}: {count}")
```

```
Object Counts:
car: 1
```

```
data = []

for box in boxes:

    class_id = int(box.cls[0])
    confidence = float(box.conf[0])

    x1, y1, x2, y2 = box.xyxy[0].tolist()

    class_name = model.names[class_id]

    data.append([
        class_name,
        confidence,
        x1,
        y1,
        x2,
        y2
    ])

df = pd.DataFrame(
    data,
    columns=[
        "Object",
        "Confidence",
```

```
        "X1",
        "Y1",
        "X2",
        "Y2"
    ]
)

df
```

	Object	Confidence	X1	Y1	X2	Y2
0	car	0.791887	1785.766602	1990.272949	6286.304199	4250.930176

```
confidence_threshold = 0.50

filtered_data = df[
    df["Confidence"] >= confidence_threshold
]

print("Detections with confidence >= 0.50")

display(filtered_data)
```

```
Detections with confidence >= 0.50
```

	Object	Confidence	X1	Y1	X2	Y2
0	car	0.791887	1785.766602	1990.272949	6286.304199	4250.930176

```
output_path = "detected_image.jpg"

cv2.imwrite(output_path, detected_image)

print("Detection result saved as:", output_path)
```

```
Detection result saved as: detected_image.jpg
```

```
from google.colab import files

files.download(output_path)
```

```
print("Number of detected objects:", len(boxes))

if len(boxes) == 0:
    print("No objects detected.")
else:
    print("\nDetected objects:")

    for obj, count in object_counts.items():
        print(f"{obj}: {count}")
```

```
Number of detected objects: 1

Detected objects:
car: 1
```

```
Start coding or generate with AI
```

Start coding or generate with AI.